

ZEROTRUST - ONE

Comprehensive Technical Architecture & Security Specification

In-depth Engineering Breakdown for Security Operations Center (SOC) & Engineering Teams

ARCH: ZERO-TRUST PIPELINE

CONSENSUS MATRIX v3.0

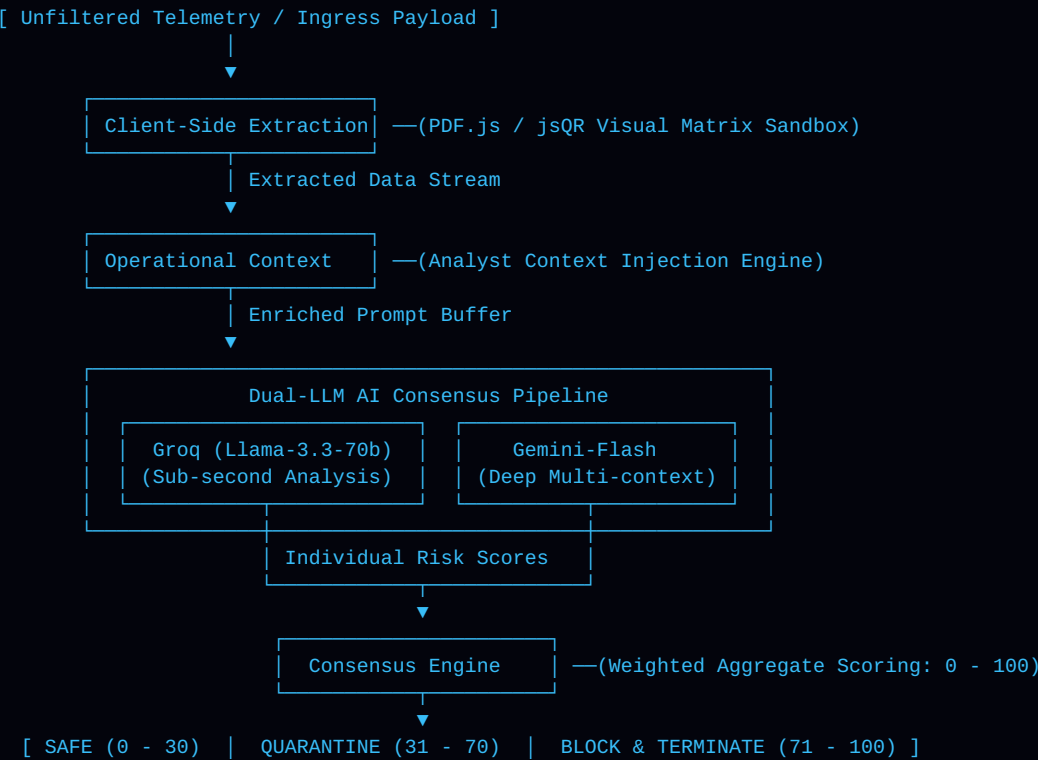
SPEC LEVEL: PROFESSIONAL

1. EXECUTIVE & ARCHITECTURAL OVERVIEW

ZeroTrust-One is an enterprise SOC telemetry inspection platform built around a strict **Never Trust, Always Verify** core model. Untrusted payloads—spanning raw email streams, document files, barcodes, and executable binaries—are strictly sandboxed, parsed, and evaluated prior to ingress into internal networks or cloud environments.

The core differentiator is its **Dual-LLM AI Consensus Matrix Engine** coupled with client-side preprocessing sandboxes. By parallel-processing payloads across specialized AI models (**Groq Llama-3.3-70b** and **Gemini-Flash**) alongside deterministic heuristic rules, ZeroTrust-One minimizes false positives while ensuring sub-second inference latency.

2. SYSTEM PIPELINE ARCHITECTURE



3. INSPECTION ENGINES & SECURITY PARSERS

A. Email Header & MIME Inspector

- **SPF Validation:** Verifies envelope sender IP against authorized publishing DNS SPF records.
- **DKIM Verification:** Checks cryptographic header signatures to ensure no in-transit tampering.

B. Client Preprocessing Sandbox

- **PDF Object Traversal:** Uses PDF.js to traverse PDF object trees locally, inspecting /JS, /JavaScript, /Launch, and /OpenAction streams.

- **DMARC Alignment:** Validates domain alignment between MIME From and envelope headers.
 - **Header Anomaly Analysis:** Flags Reply-To mismatches, hop chain anomalies, and typosquatted domains.
- **Quishing Protection:** Employs jsQR matrix decoders to extract URLs from 2D barcodes without initiating outbound browser connections.
 - **Static Binary Telemetry:** Evaluates entropy levels, section characteristics, and header signatures on raw binary files.

4. SPECIALIZED PAYLOAD MATRIX

PAYLOAD TYPE	PARSER ENGINE	SECURITY INSPECTION SCOPE
RAW_EMAIL_HEADERS_AND_BODY	Strict Security Zero-Trust	Parses MIME/RFC822 structures, SPF/DKIM/DMARC records, header anomalies, reply-to mismatches, and phishing syntax.
HTTP_URL_STRING	URI Inspector Engine	Evaluates URIs for typosquatting, IDN homograph attacks, query string parameters, redirect chains, and domain reputation.
UNSTRUCTURED_TEXT_PAYLOAD	NLP Heuristic Scanner	Analyzes free-form text buffers for social engineering, credential harvesting prompts, PII leakage, and prompt injection/jailbreaks.
QR_CODE_IMAGE_SCAN	jsQR Visual Matrix	Decodes matrix images to extract embedded URLs locally, mitigating quishing exploits without rendering remote assets.
PDF_DOCUMENT_PAYLOAD	PDF.js Extractor	Extracts text layers and structural object graphs from PDFs to detect malicious JS streams, launch triggers, and hidden streams.
BINARY_FILE_ARTIFACT	Static Artifact Analyzer	Static telemetry extraction analyzing entropy, header signatures, structural section flags, and PE/ELF markers.

5. ANALYST CONTEXT INJECTION ENGINE

Context Enrichment Feature: Standard AI security scanners evaluate payloads in complete isolation, leading to false positives on administrative scripts or custom internal emails. ZeroTrust-One allows SOC analysts to supply operational context (e.g., *"Expected vendor payment update email from HR"*). This context is injected into the prompt system message. The LLMs verify if the **declared context matches the technical telemetry signatures**, dramatically lowering false positives and catching targeted spear-phishing attempts.

6. DUAL-LLM CONSENSUS MATRIX & SCORING LOGIC

ZeroTrust-One dispatches normalized payload streams simultaneously to two complementary LLMs:

ENGINE NODE	MODEL NAME	SPECIALIZATION / ROLE	SCORE FORMAT
Groq Node	Llama-3.3-70b	High-throughput, sub-second structural pattern recognition & token risk analysis.	0 / 100
Gemini Node	Gemini-Flash	Deep contextual reasoning, intent verification, and multi-modal context alignment.	0 / 100

Verdict Thresholds & Automated Policy Enforcement

- **0 – 30 (LOW / SAFE):** Payload passed all structural, heuristic, and AI consensus checks. Enforcement: **NO THREAT DETECTED**.
- **31 – 70 (MEDIUM / SUSPICIOUS):** Anomaly detected or model divergence. Enforcement: **QUARANTINE & SOC REVIEW**.
- **71 – 100 (HIGH / CRITICAL):** Active threat vector confirmed (e.g., spoofed auth + credential harvesting URI). Enforcement: **BLOCK & TERMINATE SESSION**.

